

2D Stationary Thermal Problems with Linear Triangular Elements

ISHIAQ Ahmed Olamide

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1 Project Overview

The aim of the project is to complete and understand a Python finite element code for two-dimensional stationary thermal problems. The main solver is provided in `solveFE.py`. The main implementation work is in `tri3ThermalDirect.py`, it includes the following:

- `computeKe`: computes the elementary stiffness matrix;
- `computeFve`: computes the elementary volume/source load vector;
- `computeFNe`: computes the elementary Neumann boundary load vector.

The project is therefore not only a coding task. It is a complete finite element workflow:

1. Define the physical heat conduction problem;
2. Derive the weak form;
3. Discretize the domain using triangular elements;
4. Compute local element matrices and vectors;
5. Assemble the global linear system;
6. Apply boundary conditions;
7. Solve for nodal temperatures;
8. Validate the numerical solution against analytical solutions;

2 Governing Equation

The physical problem is steady-state heat conduction in a two-dimensional domain Ω :

$$-\nabla \cdot (K \nabla u) = r \quad \text{in } \Omega, \quad (1)$$

where:

- $u(x, y)$ is the unknown temperature field;
- K is the isotropic thermal conductivity;
- r is a volumetric heat source term.

The heat flux vector is defined as

$$\mathbf{q} = -K \nabla u. \quad (2)$$

The boundary $\partial\Omega$ is split into two main parts:

$$\partial\Omega = \Gamma_D \cup \Gamma_N, \quad \Gamma_D \cap \Gamma_N = \emptyset. \quad (3)$$

On the Dirichlet boundary, the temperature is prescribed:

$$u = u_D \quad \text{on } \Gamma_D. \quad (4)$$

On the Neumann boundary, a flux-type condition is prescribed.

$$K \nabla u \cdot \mathbf{n} = q_N \quad \text{on } \Gamma_N, \quad (5)$$

so that the Neumann contribution appears with a positive sign in the weak form. The physical outward heat flux is $\mathbf{q} \cdot \mathbf{n}$, then $\mathbf{q} \cdot \mathbf{n} = -K \nabla u \cdot \mathbf{n}$, and the sign of q_N is adjusted accordingly.

3 Weak Form

Let v be a test function that vanishes on the Dirichlet boundary Γ_D . Multiplying Equation (1) by v and integrating over Ω gives

$$\int_{\Omega} -v \nabla \cdot (K \nabla u) d\Omega = \int_{\Omega} v r d\Omega. \quad (6)$$

After integration by parts, the weak form becomes

$$\int_{\Omega} K \nabla v \cdot \nabla u d\Omega = \int_{\Omega} v r d\Omega + \int_{\Gamma_N} v q_N d\Gamma, \quad (7)$$

provided the Neumann convention in Equation (5) is used.

The finite element method converts Equation (7) into the algebraic system

$$\mathbf{K}\mathbf{U} = \mathbf{F}, \quad (8)$$

where:

- $\mathbf{K}$ is the global stiffness matrix;
- $\mathbf{U}$ is the vector of unknown nodal temperatures;
- $\mathbf{F}$ is the global load vector.

4 Project File Roles

Table 1: Main project files and their roles.

File	Purpose
testFE.py	Defines the problem: mesh name, conductivity, source term, boundary conditions, and output name.
solveFE.py	Main finite element solver: reads the mesh, assembles the global system, applies boundary conditions, and solves the linear system.
tri3ThermalDirect.py	Computes element stiffness matrices and element load vectors for 3-node triangular thermal elements.
gmshParser.py	Reads Gmsh .msh mesh files.
integrationRule.py	Provides numerical integration points and weights.
export.py	Exports results, typically to Gmsh .pos format.
*.msh files	Mesh files generated using Gmsh.
*.geo files	Geometry files used to generate meshes in Gmsh.

5 The 3-Node Linear Triangular Element

The element used in the project is the 3-node linear triangular element, often called a T3 element. For one triangular element Ω_e with nodes

$$(x_1, y_1), \quad (x_2, y_2), \quad (x_3, y_3), \quad (9)$$

the approximate temperature field is

$$u_h(x, y) = N_1(x, y)u_1 + N_2(x, y)u_2 + N_3(x, y)u_3, \quad (10)$$

or, in matrix form,

$$u_h = \mathbf{N} \mathbf{u}_e, \quad \mathbf{N} = [N_1 \quad N_2 \quad N_3], \quad \mathbf{u}_e = [u_1 \quad u_2 \quad u_3]^T. \quad (11)$$

Because the shape functions are linear, their derivatives are constant inside the element. Therefore, the gradient of the temperature field can be written as

$$\nabla u_h = \mathbf{B} \mathbf{u}_e, \quad (12)$$

where $\mathbf{B}$ is the strain-gradient matrix for the thermal problem.

6 Element Geometry

The signed area of the triangle is

$$A_s = \frac{1}{2} \begin{vmatrix} 1 & x_1 & y_1 \\ 1 & x_2 & y_2 \\ 1 & x_3 & y_3 \end{vmatrix}. \quad (13)$$

In the code, the physical area should be taken as

$$A_e = |A_s|. \quad (14)$$

Define the geometric coefficients

$$b_1 = y_2 - y_3, \quad b_2 = y_3 - y_1, \quad b_3 = y_1 - y_2, \quad (15)$$

$$c_1 = x_3 - x_2, \quad c_2 = x_1 - x_3, \quad c_3 = x_2 - x_1. \quad (16)$$

The matrix $\mathbf{B}$ is then

$$\mathbf{B} = \frac{1}{2A_e} \begin{bmatrix} b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{bmatrix}. \quad (17)$$

7 Element Stiffness Matrix

The elementary stiffness matrix is

$$\mathbf{K}_e = \int_{\Omega_e} K \mathbf{B}^T \mathbf{B} d\Omega. \quad (18)$$

For a T3 element with constant isotropic conductivity K , $\mathbf{B}$ is constant. Therefore,

$$\mathbf{K}_e = K A_e \mathbf{B}^T \mathbf{B}. \quad (19)$$

7.1 Reference Check

For the reference triangle

$$(0, 0), \quad (1, 0), \quad (0, 1), \quad (20)$$

with $K = 1$, the expected stiffness matrix is

$$\mathbf{K}_e = \begin{bmatrix} 1 & -0.5 & -0.5 \\ -0.5 & 0.5 & 0 \\ -0.5 & 0 & 0.5 \end{bmatrix}. \quad (21)$$

8 Element Volume/Source Vector

The elementary volume/source load vector is

$$\mathbf{F}_{v,e} = \int_{\Omega_e} \mathbf{N}^T r \, d\Omega. \quad (22)$$

If r is constant over the element, then

$$\mathbf{F}_{v,e} = \frac{r A_e}{3} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}. \quad (23)$$

For a triangle with area $A_e = 0.5$ and constant source $r = 1$, the expected source vector is

$$\mathbf{F}_{v,e} = \frac{0.5}{3} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1/6 \\ 1/6 \\ 1/6 \end{bmatrix}. \quad (24)$$

9 Element Neumann Boundary Vector

The elementary Neumann load vector on a boundary edge Γ_e is

$$\mathbf{F}_{N,e} = \int_{\Gamma_e} \mathbf{N}^T q_N \, d\Gamma. \quad (25)$$

For a 2-node boundary edge with constant flux q_N , this becomes

$$\mathbf{F}_{N,e} = \frac{q_N L_e}{2} \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad (26)$$

where L_e is the edge length.

For an edge of length $L_e = 1$ and flux $q_N = 1$, the expected vector is

$$\mathbf{F}_{N,e} = \begin{bmatrix} 0.5 \\ 0.5 \end{bmatrix}. \quad (27)$$

10 Analytical Validation Problem 1: Linear Solution

A useful validation case is a square domain with:

$$u = 0 \quad \text{on the bottom boundary,} \quad (28)$$

$$q_N = 1 \quad \text{on the top boundary,} \quad (29)$$

$$r = 0, \quad (30)$$

$$K = 1. \quad (31)$$

The exact solution is linear:

$$u(x, y) = y, \quad (32)$$

assuming the square extends from $y = 0$ to $y = 1$.

More generally, if the bottom coordinate is y_0 , then

$$u(x, y) = \frac{q_N}{K} (y - y_0). \quad (33)$$

This is a strong validation test because a linear triangular element should reproduce a linear solution almost exactly.

11 Analytical Validation Problem 2: Square with Unit Source

The second validation problem is defined on

$$(x, y) \in [-1, 1] \times [-1, 1]. \quad (34)$$

The boundary and source conditions are

$$u = 0 \quad \text{on all boundaries,} \quad (35)$$

$$K = 1, \quad (36)$$

$$r = 1. \quad (37)$$

The exact solution is expressed as a double sine series:

$$u(x, y) = \sum_{i=1,3,5,\dots} \sum_{j=1,3,5,\dots} \frac{64}{\pi^4(i^2 + j^2)ij} \sin\left(\frac{i\pi(x+1)}{2}\right) \sin\left(\frac{j\pi(y+1)}{2}\right). \quad (38)$$

12 Conclusion

This project is a compact but complete example of the finite element method applied to stationary heat conduction. The key mathematical ideas are the weak form, the interpolation of the T3 element, the stiffness matrix $\mathbf{K}_e = \mathbf{K} \mathbf{A}_e \mathbf{B}^T \mathbf{B}$, and the correct assembly of the source and Neumann load vectors.